

Data Cleaning, Tidying and Visualization: Sustainability Survey

AUTHOR

Dung (Tony) Le

PUBLISHED

March 18, 2026

Overview

This document cleans and tidies the Fall 2015 Truman State University Sustainability Survey dataset. Column renaming was completed in Excel (see the Dictionary tab) and the working sheet was exported as [Working.csv](#). All remaining cleaning steps are performed below in R. My goal in this assignment is to make a subset of this data set as clean and tidy as possible.

Question 2

Setup: Load Libraries and Data

```
# Load the tidyverse package
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4      ✓ readr      2.1.6
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.1      ✓ tibble     3.3.1
✓ lubridate  1.9.4      ✓ tidyr      1.3.2
✓ purrr      1.2.1

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Read the CSV that was exported from the Working sheet in Excel.
# na.strings converts both empty strings and "NA" text into proper R NA values.
df <- read.csv("../raw_data/Working.csv",
               stringsAsFactors = FALSE,
               na.strings       = c("", "NA"))

# glimpse() gives a compact overview: number of rows/columns,
# column names, data types, and first few values of each column.
glimpse(df)
```

Rows: 182

Columns: 35

```

$ ID                <int> 253940, 253941, 253958, 254012, 254013, 254037, 25407...
$ Gender            <chr> "Other/ prefer not to say", "Male", "Male", "Female",...
$ Bmonth            <int> 6, 2, 9, 5, 6, 8, 1, 1, 7, 10, 9, 8, 4, 10, 5, 7, 6, ...
$ Byear             <int> 1994, 1995, 1995, 1996, 1993, 1997, 1997, 1997, 1995,...
$ Grade             <chr> "Third-year", "First-year ", "First-year ", "Third-ye...
$ School            <chr> "HSE SAM BUS", NA, NA, "SCS", "SCS", "SCS", "SCS", "S...
$ Confidence        <chr> "Neutral", "Not Confident at All", "Not Confident at ...
$ Heard_sustain     <chr> NA, NA, NA, "Sustainability Week", NA, "Sustainabilit...
$ Heard_earthweek   <chr> NA, NA, NA, "Earth Week", "Earth Week", "Earth Week",...
$ Heard_greenthumb  <chr> "Green Thumb", NA, NA, NA, NA, NA, NA, NA, "Green Thu...
$ Heard_bike        <chr> "Bike Co-op", NA, NA, "Bike Co-op", "Bike Co-op", "Bi...
$ Heard_conference  <chr> "ENVS Conference", NA, NA, NA, "ENVS Conference", NA,...
$ Heard_recycle     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ Heard_buddies     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...
$ Part_sustain      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,...
$ Part_earthweek    <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ Part_greenthumb   <int> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,...
$ Part_bike         <int> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,...
$ Part_conference   <int> 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,...
$ Part_recycle      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ Part_buddies      <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,...
$ Effective         <chr> "Ineffective", NA, NA, NA, NA, NA, "Neutrale", NA, "N...
$ More_activities   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Glass recycl...
$ Attendance        <int> NA, 1, 1, 1, 4, 3, 2, 3, 2, 4, 3, 5, 1, 3, 2, 4, 3, 2...
$ Collaborate       <int> NA, 1, 1, 1, 3, 4, 2, 3, 3, 4, 3, 5, 3, 3, 2, 5, 3, 3...
$ Member            <int> NA, 1, 1, 1, 4, 3, 2, 3, 3, 4, 4, 3, 1, 4, 2, 4, 3, 4...
$ Office_space      <chr> "No preference", "No preference", "Physical Office", ...
$ Increase_rate     <chr> "Not willing at all", "Not willing at all", "Not will...
$ Work_sustain      <chr> "no", "no", "no", "no", "yes", "no", "no", "yes", "no...
$ Leadership        <int> NA, NA, NA, NA, 4, NA, NA, 3, NA, 4, 4, 1, NA, NA, NA...
$ Manage_funds      <int> NA, NA, NA, NA, 2, NA, NA, 3, NA, 4, 4, 1, NA, NA, NA...
$ Investigate       <int> NA, NA, NA, NA, 4, NA, NA, 3, NA, 4, 4, 3, NA, NA, NA...
$ Solutions         <int> NA, NA, NA, NA, 5, NA, NA, 3, NA, 4, 4, 4, NA, NA, NA...
$ Brainstorm        <int> NA, NA, NA, NA, 5, NA, NA, 3, NA, 4, 3, 5, NA, NA, NA...
$ Comments          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, N...

```

Data Cleaning Steps

(a) Recode Gender as a Factor with 3 Short Categories

```

# The raw data contains three character values for Gender.
# We recode them to short labels and convert to a factor so that
# R treats Gender as a categorical variable (not free text) in
# all plots and models.

```

```
df <- df %>%
  mutate(
    Gender = case_when(
      Gender == "Female" ~ "Female", # keep as-is
      Gender == "Male" ~ "Male", # keep as-is
      Gender == "Other/ prefer not to say" ~ "Other", # shorten the label
      TRUE ~ NA_character_ # any unexpected value -> NA
    ),
    # Wrap the recoded character column in factor() with explicit levels.
    # This controls the display order in every table and graph.
    Gender = factor(Gender, levels = c("Female", "Male", "Other"))
  )

# Verify: should show Female=140, Male=38, Other=4
df %>% count(Gender)
```

```
Gender    n
1 Female 140
2  Male   38
3  Other   4
```

(b) Recode Grade Level to Numeric Values 1–5

Step 1: trimws() strips any leading or trailing whitespace from the
 # Grade column. Without this, "First-year " (with a hidden trailing space)
 # would not match "First-year" in case_when and would silently become NA.
 # Step 2: case_when maps each year-description to an integer 1–5 so that
 # graphs automatically display grade levels in the correct chronological order.

```
df <- df %>%
  mutate(
    Grade = trimws(Grade), # remove hidden whitespace (e.g. "First-year ")
    Grade = case_when(
      Grade == "First-year" ~ 1,
      Grade == "Second-year" ~ 2,
      Grade == "Third-year" ~ 3,
      Grade == "Fourth-year" ~ 4,
      Grade == "more than 4 years at Truman" ~ 5,
      TRUE ~ NA_real_ # safety net for unexpected values
    )
  )

# Verify: should show counts for integers 1–5, no NAs
df %>% count(Grade)
```

```
Grade    n
1      1 55
```

```
2      2 40
3      3 45
4      4 40
5      5  2
```

Explanation: The homework warns that some grade-level values contain extra spaces. `trimws()` solves this in one step rather than having to hard-code every possible spacing variant inside `case_when`. The result is numeric so R can sort and scale grade levels correctly on plot axes.

(c) Split School Column into School1, School2, and School3

```
# The School column holds one, two, or three school abbreviations
# separated by single spaces (e.g. "HSE SAM BUS").
# separate() splits the column on whitespace into up to three pieces.
#   extra = "drop" -> silently discard any token beyond the 3rd
#   fill  = "right" -> pad with NA on the right for students with
#                       fewer than 3 schools listed
# Note: we pass col as a quoted string "School" to avoid the
# "object 'School' not found" error that occurs in some R versions
# when the bare column name is used.

df <- df %>%
  separate(
    col   = "School",
    into  = c("School1", "School2", "School3"),
    sep   = " ",
    extra = "drop",    # ignore any extra tokens beyond 3
    fill  = "right"    # fill unused columns with NA
  )

# Verify the split worked as expected
cat("Students listing 2+ schools:", sum(!is.na(df$School2)), "\n")
```

Students listing 2+ schools: 22

```
cat("Students listing 3 schools:", sum(!is.na(df$School3)), "\n")
```

Students listing 3 schools: 1

```
df %>% select(School1, School2, School3) %>% head(10)
```

	School1	School2	School3
1	HSE	SAM	BUS
2	<NA>	<NA>	<NA>
3	<NA>	<NA>	<NA>
4	SCS	<NA>	<NA>
5	SCS	<NA>	<NA>

6	SCS	<NA>	<NA>
7	SCS	<NA>	<NA>
8	SCS	<NA>	<NA>
9	SCS	SAL	<NA>
10	HSE	<NA>	<NA>

(d) Create the SAM Binary Indicator Column

```
# Create a column named SAM that equals 1 if the student
# listed the School of Science and Mathematics in ANY of the three school
# columns, and 0 otherwise.
#
# if_else() evaluates the logical condition for each row:
#   - TRUE  -> 1L (the student listed SAM)
#   - FALSE -> 0L (the student did not list SAM)
#   - missing = 0L -> if a comparison involves NA (e.g. School2 is NA),
#                       treat it as "not SAM" rather than producing NA

df <- df %>%
  mutate(
    SAM = if_else(
      School1 == "SAM" | School2 == "SAM" | School3 == "SAM",
      true      = 1L,
      false     = 0L,
      missing   = 0L # NA comparisons treated as FALSE (not listed = not SAM)
    )
  )

# Verify: total should be 50 SAM students
df %>% count(SAM)
```

	SAM	n
1	0	132
2	1	50

(e) Recode Heard_sustain from Text to 0/1

```
# The "heard about" columns store a text label (e.g. "Sustainability Week")
# when the student checked that activity, and NA when they did not.
# The "participated" columns use 1 (participated) and 0 (did not).
# We need to make Heard_sustain consistent with the participated coding.
#
# case_when maps:
#   "Sustainability Week" -> 1 (heard about it)
#   anything else / NA    -> 0 (did not hear about it)
```

```
# Show the values BEFORE recoding so the change is visible in the PDF
cat("---- Before recoding ----\n")
```

---- Before recoding ----

```
df %>% count(Heard_sustain) %>% print()
```

```
      Heard_sustain    n
1 Sustainability Week 105
2                <NA>  77
```

```
df <- df %>%
  mutate(
    Heard_sustain = case_when(
      Heard_sustain == "Sustainability Week" ~ 1L, # heard = 1
      TRUE ~ 0L # not heard (including NA) = 0
    )
  )
```

```
# Show the values AFTER recoding to confirm the transformation
cat("\n---- After recoding ----\n")
```

---- After recoding ----

```
df %>% count(Heard_sustain) %>% print()
```

```
      Heard_sustain    n
1                0  77
2                1 105
```

(f) Fix Typos in the Program Effectiveness Variable

```
# The Effective column contains a clear data-entry typo: "Neutrale" instead
# of "Neutral". str_replace() with anchors (^ = start, $ = end) matches only
# the exact misspelling and replaces it, leaving all other values untouched.
```

```
cat("---- Before fixing typos ----\n")
```

---- Before fixing typos ----

```
df %>% count(Effective) %>% print()
```

```
      Effective    n
1 Completely Effective  6
```

```

2 Completely Ineffective 4
3           Effective 25
4           Ineffective 12
5           Neutral 7
6           Neutrale 31
7           <NA> 97

```

```

df <- df %>%
  mutate(
    Effective = str_replace(Effective, "^Neutrale$", "Neutral")
    # ^ anchors to start of string, $ anchors to end,
    # so only the exact word "Neutrale" is replaced.
  )

cat("\n--- After fixing typos ---\n")

```

--- After fixing typos ---

```
df %>% count(Effective) %>% print()
```

```

           Effective  n
1  Completely Effective 6
2  Completely Ineffective 4
3           Effective 25
4           Ineffective 12
5           Neutral 38
6           <NA> 97

```

Explanation: Using anchors `^` and `$` is a safety measure — it prevents accidentally replacing “Neutrale” if it appeared as part of a longer phrase. After this step, “Neutral” should show a combined count of 38 (31 + 7 from the original dirty data).

(g) Recode Tuition Increase as an Ordered Factor

```

# The Increase_rate column is currently a character variable.
# By default R would display these values in alphabetical order, which is
# meaningless for this ordinal variable ("1-10" before "Not willing at all"
# alphabetically, for example).
#
# Solution: convert to an ORDERED factor with the five levels listed in the
# exact sequence specified in the assignment. R will then display and sort
# these values correctly in every plot and table automatically.
#
# IMPORTANT: the level labels below match the EXACT capitalisation found in
# the raw data – do NOT change them. Mismatched capitalisation would cause
# all values to become NA.

```

```
# Show current (alphabetical / arbitrary) ordering
cat("---- Before recoding ----\n")
```

```
---- Before recoding ----
```

```
df %>% count(Increase_rate) %>% print()
```

```
      Increase_rate  n
1          $1-$10  80
2          $11-$20  29
3          $21-$30  16
4 Not willing at all  50
5 anything above $30   7
```

```
# Define the correct ordinal sequence (low to high willingness)
```

```
tuition_levels <- c(
  "Not willing at all", # least willing
  "$1-$10",
  "$11-$20",
  "$21-$30",
  "anything above $30" # most willing
)
```

```
df <- df %>%
  mutate(
    Increase_rate = factor(
      Increase_rate,
      levels = tuition_levels,
      ordered = TRUE # ordered = TRUE marks this as an ordinal variable
    )
  )
```

```
# Verify: levels should now appear in the correct low-to-high order
```

```
cat("\n--- After recoding - correct ordinal levels ---\n")
```

```
--- After recoding - correct ordinal levels ---
```

```
print(levels(df$Increase_rate))
```

```
[1] "Not willing at all" "$1-$10"          "$11-$20"
[4] "$21-$30"           "anything above $30"
```

```
df %>% count(Increase_rate) %>% print()
```

```
      Increase_rate  n
1 Not willing at all  50
```


2	\$1–\$10	80
3	\$11–\$20	29
4	\$21–\$30	16
5	anything above \$30	7

Explanation: `ordered = TRUE` tells R that the categories have a meaningful rank (`"Not willing at all" < "$1–$10" < ... < "anything above $30"`). Without it, R treats the factor as nominal and any graph would display bars or axes in an arbitrary or alphabetical order. We do **not** apply `str_to_title()` because the raw data already uses the exact same capitalisation as the assignment's specified labels — changing case would cause non-matching values to silently become `NA`.

(h) Export the Cleaned Dataset

```
# write.csv exports the cleaned dataframe to a CSV file.
# row.names = FALSE prevents R from adding an unwanted row-number column
# as the first column of the output file.

write.csv(df, "sustainability_cleaned_TonyLe.csv", row.names = FALSE)

# Confirm the export succeeded and show the final dimensions
cat("Export complete:", nrow(df), "rows x", ncol(df), "columns.\n")
```

Export complete: 182 rows x 38 columns.

```
cat("File saved as: sustainability_cleaned_TonyLe.csv\n")
```

File saved as: sustainability_cleaned_TonyLe.csv

Question 3

Visualization: Tuition Willingness by First-Listed School

Graph

```
# Lookup table: map short abbreviations to full school names for readable labels
school_labels <- c(
  BUS = "Business",
  HSE = "Health Sci & Ed",
  IDS = "Undeclared/IDS",
  SAL = "Arts & Letters",
  SAM = "Science & Math",
  SCS = "Social & Cultural"
```

```

)

# Colour palette: red = low willingness, blue = high willingness
# (intuitive: warm colours = reluctant, cool colours = willing)
will_colors <- c(
  "Not willing at all" = "#D73027", # deep red
  "$1-$10"            = "#F46D43", # orange-red
  "$11-$20"           = "#FEE090", # yellow
  "$21-$30"           = "#74ADD1", # light blue
  "anything above $30" = "#313695"  # deep blue
)

# Step 1: Compute mean willingness score per school so we can order schools
#         from least to most willing on the x-axis.
#         as.integer() converts the ordered factor levels to 1-5 numerically.
school_order <- df %>%
  filter(!is.na(School1), !is.na(Increase_rate)) %>%
  mutate(will_num = as.integer(Increase_rate)) %>%
  group_by(School1) %>%
  summarise(mean_will = mean(will_num), .groups = "drop") %>%
  arrange(mean_will) %>%
  mutate(School_Full = recode(School1, !!!school_labels))

# Step 2: Build the percentage table for the stacked bars
plot_df <- df %>%
  filter(!is.na(School1), !is.na(Increase_rate)) %>%
  mutate(School_Full = recode(School1, !!!school_labels)) %>%
  count(School_Full, Increase_rate) %>%
  group_by(School_Full) %>%
  mutate(pct = n / sum(n) * 100) %>% # convert counts to percentages within each school
  ungroup() %>%
  # Apply the school ordering determined in Step 1
  mutate(School_Full = factor(School_Full,
                              levels = school_order$School_Full))

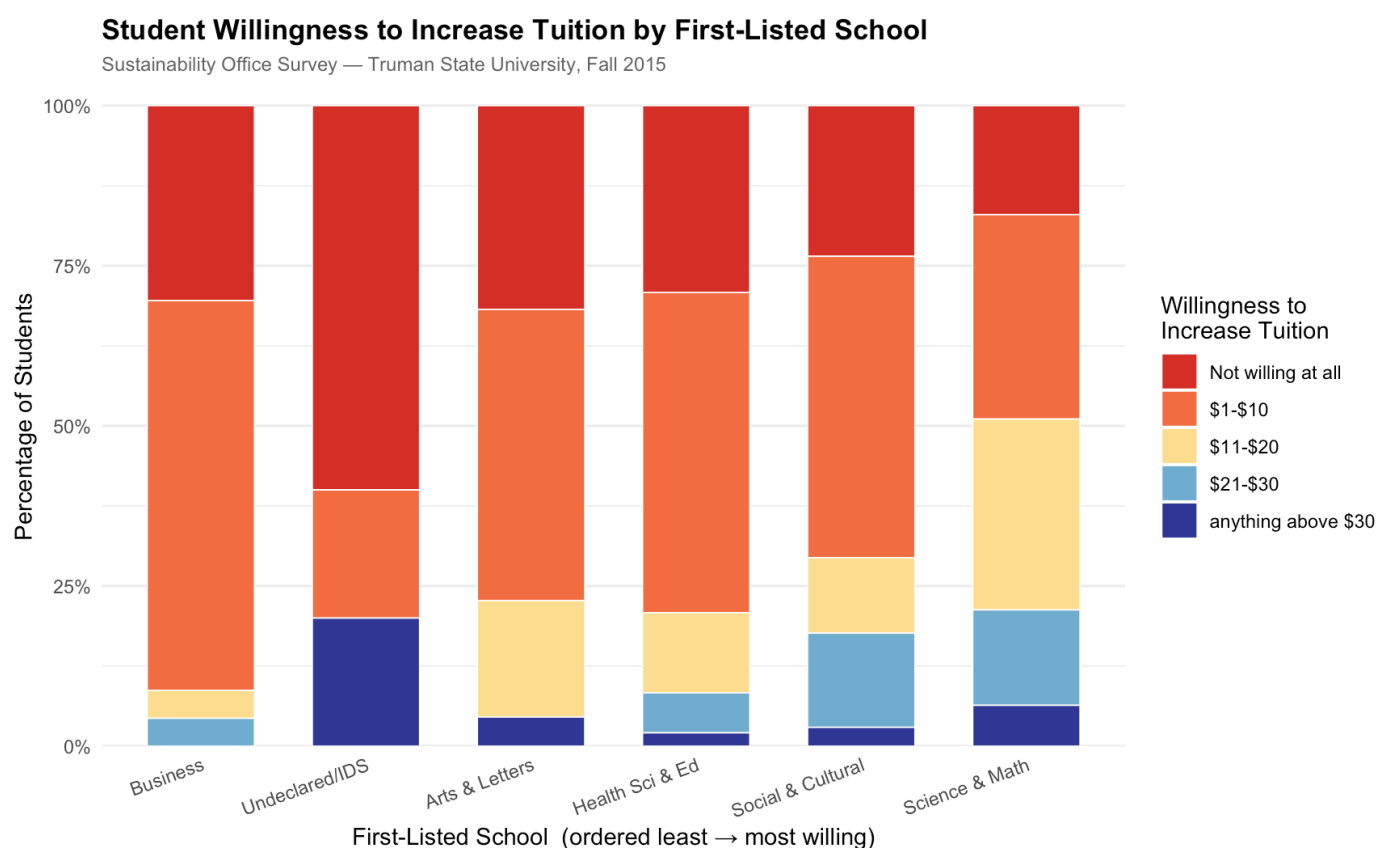
# Step 3: Draw the stacked bar chart
ggplot(plot_df,
  aes(x = School_Full,
      y = pct,
      fill = Increase_rate)) +
  geom_col(position = "stack",
    width = 0.65,
    colour = "white", # thin white borders separate the stacked segments
    linewidth = 0.3) +
  scale_fill_manual(
    values = will_colors,
    name = "Willingness to\nIncrease Tuition"
  ) +
  scale_y_continuous(
    labels = scales::label_percent(scale = 1), # format y-axis as percentages
    expand = expansion(mult = c(0, 0.03))
  )

```

```

) +
labs(
  title    = "Student Willingness to Increase Tuition by First-Listed School",
  subtitle = "Sustainability Office Survey – Truman State University, Fall 2015",
  x        = "First-Listed School (ordered least → most willing)",
  y        = "Percentage of Students"
) +
theme_minimal(base_size = 11) +
theme(
  plot.title       = element_text(face = "bold", size = 13),
  plot.subtitle    = element_text(colour = "grey40", size = 9),
  axis.text.x      = element_text(angle = 20, hjust = 1),
  panel.grid.major.x = element_blank(),
  legend.position  = "right"
)

```



Stacked bar chart showing the percentage distribution of tuition-increase willingness for each first-listed school. Red shades = less willing, blue shades = more willing. Schools are ordered from lowest to highest mean willingness (left to right).

Summary Table

```

# Compute the key statistics per school needed to answer Q3b.
# will_num converts the ordered factor to an integer 1-5 for arithmetic.
summary_tbl <- df %>%

```

```

filter(!is.na(School1), !is.na(Increase_rate)) %>%
mutate(
  will_num = as.integer(Increase_rate),
  School_Full = recode(School1, !!!school_labels)
) %>%
group_by(School_Full) %>%
summarise(
  N = n(),
  Mean_Willingness = round(mean(will_num), 2), # average ordinal score 1-5
  Pct_Not_Willing = round(mean(will_num == 1) * 100, 1), # % who said "Not willing at
  .groups = "drop"
) %>%
arrange(Mean_Willingness) # sort from least to most willing

knitr::kable(
  summary_tbl,
  col.names = c("School", "N", "Mean Willingness (1-5)",
                "% Not Willing At All"),
  caption = paste("Summary of tuition-increase willingness by first-listed school.",
                  "Scale: 1 = Not willing at all, 5 = Anything above $30.",
                  "Sorted from least to most willing.")
)

```

Summary of tuition-increase willingness by first-listed school. Scale: 1 = Not willing at all, 5 = Anything above \$30. Sorted from least to most willing.

School	N	Mean Willingness (1–5)	% Not Willing At All
Business	23	1.83	30.4
Arts & Letters	22	2.00	31.8
Undeclared/IDS	5	2.00	60.0
Health Sci & Ed	48	2.02	29.2
Social & Cultural	34	2.26	23.5
Science & Math	47	2.62	17.0

Interpretation (Q3b)

```

# Extract the least and most willing schools programmatically
least_willing <- summary_tbl %>% slice_min(Mean_Willingness, n = 1)
most_willing <- summary_tbl %>% slice_max(Mean_Willingness, n = 1)

cat("School with LEAST willingness:", least_willing$School_Full,
    "(mean =", least_willing$Mean_Willingness, ")\n")

```

School with LEAST willingness: Business (mean = 1.83)

```
cat("School with MOST willingness:", most_willing$School_Full,  
    "(mean =", most_willing$Mean_Willingness, ")\n")
```

School with MOST willingness: Science & Math (mean = 2.62)

Answer to Q3b:

- The school with the **least** willingness to increase tuition is **Business** (mean score = 1.83, 30.4% "Not willing at all") — appearing at the **top** of the table above since it is sorted low-to-high.
- The school with the **most** willingness is **Science & Math** (mean score = 2.62, only 17.0% "Not willing at all") — appearing at the **bottom** of the table.
- Across all schools, the two lowest categories ("Not willing at all" and "\$1-\$10") account for the majority of responses in every group, suggesting that cost resistance is a university-wide phenomenon and not driven by any one school in particular.

```
library(plotly)    # interactive plots
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

The following object is masked from 'package:stats':

filter

The following object is masked from 'package:graphics':

layout

```
library(scales)    # format axis labels
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
df <- read.csv("../raw_data/Working.csv",  
               stringsAsFactors = FALSE,
```

```
na.strings = c("", "NA"))
```

Exploratory Data Analysis

1. Gender & Grade Level Distribution

```
# --- Plot 1: Gender distribution (interactive pie chart) ---
gender_counts <- df %>%
  mutate(Gender = case_when(
    Gender == "Female" ~ "Female",
    Gender == "Male" ~ "Male",
    Gender == "Other/ prefer not to say" ~ "Other"
  )) %>%
  count(Gender)

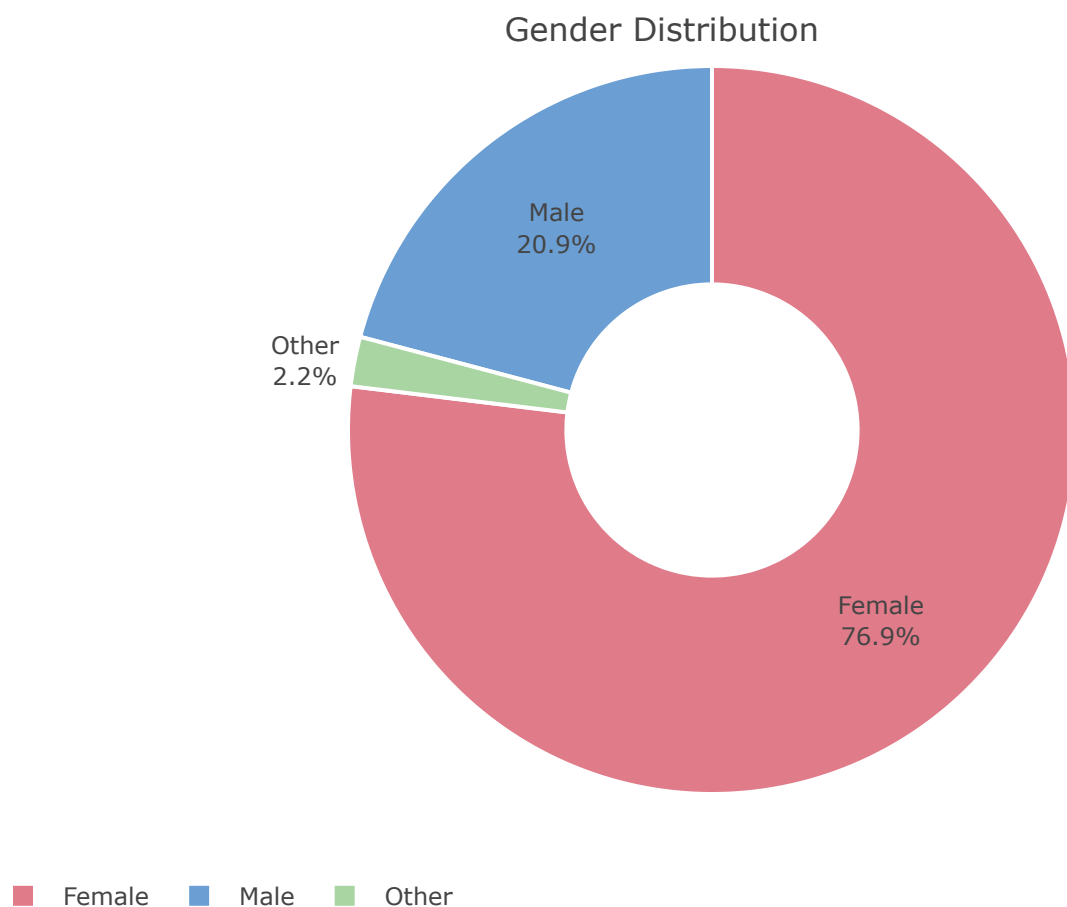
p1 <- plot_ly(
  gender_counts,
  labels = ~Gender,
  values = ~n,
  type = "pie",
  hole = 0.4, # donut style
  marker = list(colors = c("#E07B8A", "#6B9FD4", "#A8D5A2"),
    line = list(color = "white", width = 2)),
  textinfo = "label+percent",
  hovertemplate = "<b>{%label}</b><br>Count: {%value}<br>{%percent}<extra></extra>"
) %>%
  layout(
    title = list(text = "Gender Distribution", font = list(size = 16)),
    showlegend = TRUE,
    legend = list(orientation = "h", y = -0.1)
  )

# --- Plot 2: Grade level distribution (interactive bar chart) ---
grade_counts <- df %>%
  mutate(
    Grade = trimws(Grade),
    Grade = case_when(
      Grade == "First-year" ~ "1st Year",
      Grade == "Second-year" ~ "2nd Year",
      Grade == "Third-year" ~ "3rd Year",
      Grade == "Fourth-year" ~ "4th Year",
      Grade == "more than 4 years at Truman" ~ "5+ Years"
    ),
    Grade = factor(Grade,
      levels = c("1st Year", "2nd Year", "3rd Year",
        "4th Year", "5+ Years"))
  ) %>%
```

```
count(Grade)

p2 <- plot_ly(
  grade_counts,
  x = ~Grade,
  y = ~n,
  type = "bar",
  marker = list(
    color = c("#D4E6F1", "#A9CCE3", "#7FB3D3", "#5499C2", "#2E86C1"),
    line = list(color = "white", width = 1.5)
  ),
  text = ~n,
  textposition = "outside",
  hovertemplate = "<b>{%x}</b><br>Count: {%y}<extra></extra>"
) %>%
  layout(
    title = list(text = "Grade Level Distribution", font = list(size = 16)),
    xaxis = list(title = "Grade Level"),
    yaxis = list(title = "Number of Students", range = c(0, 65)),
    plot_bgcolor = "rgba(0,0,0,0)",
    paper_bgcolor = "rgba(0,0,0,0)"
  )

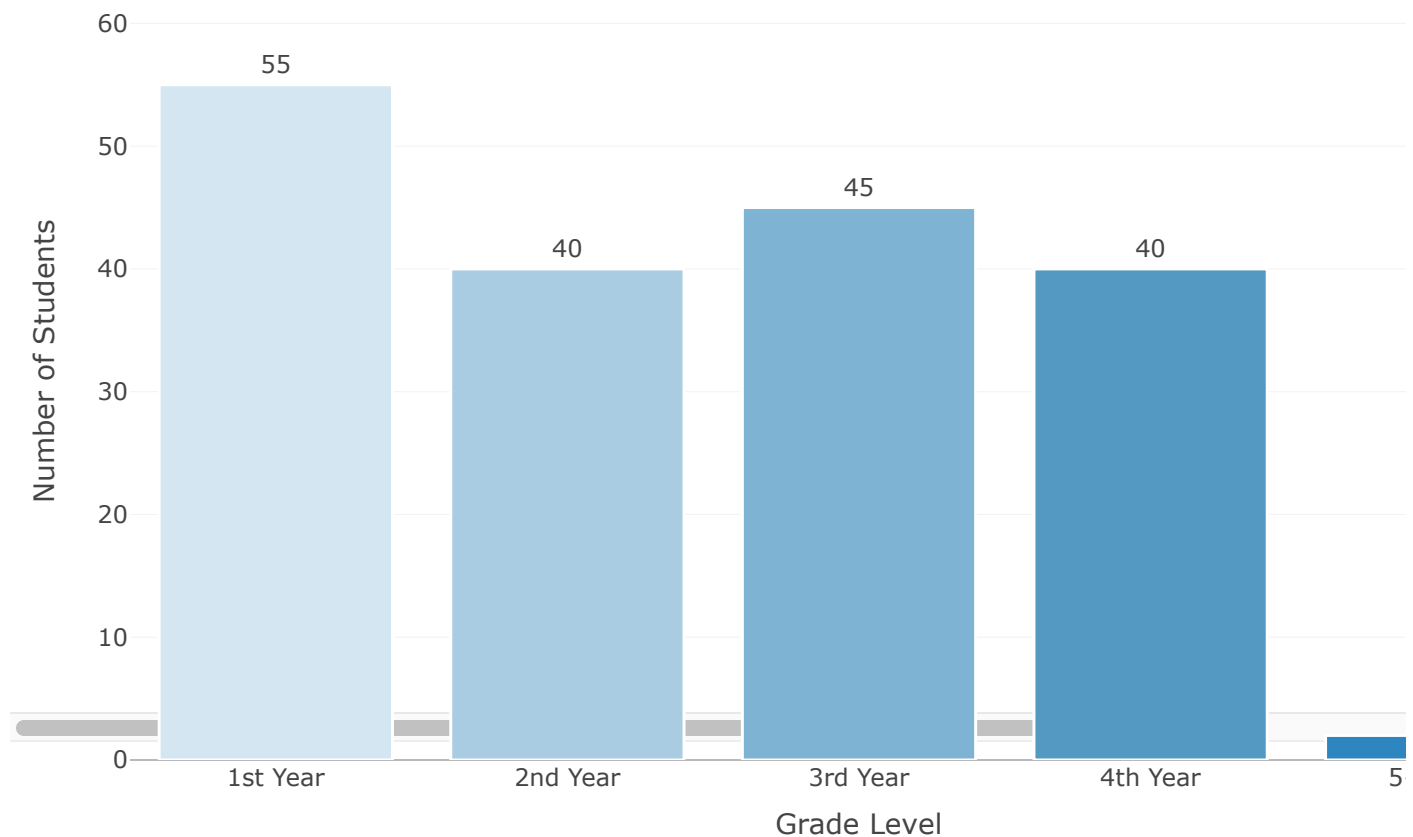
# Display both plots
p1
```



Distribution of survey respondents by Gender and Grade Level

p2

Grade Level Distribution



Distribution of survey respondents by Gender and Grade Level

2. Heard About vs. Participated in Activities

```
# Recode all Heard columns to 0/1 first for this plot
df_heard <- df %>%
  mutate(across(starts_with("Heard_"),
    ~ if_else(!is.na(.), 1L, 0L)))

# Activity names (short labels)
activity_labels <- c(
  "Heard_sustain" = "Sustainability Week",
  "Heard_earthweek" = "Earth Week",
  "Heard_greenthumb" = "Green Thumb",
  "Heard_bike" = "Bike Co-op",
  "Heard_conference" = "ENVS Conference",
```



```

    "Heard_recycle"      = "Recyclemania",
    "Heard_buddies"      = "ENV Buddies"
  )

part_labels <- c(
  "Part_sustain"        = "Sustainability Week",
  "Part_earthweek"      = "Earth Week",
  "Part_greenthumb"     = "Green Thumb",
  "Part_bike"           = "Bike Co-op",
  "Part_conference"     = "ENVS Conference",
  "Part_recycle"        = "Recyclemania",
  "Part_buddies"        = "ENV Buddies"
)

# Build summary table
heard_sums <- df_heard %>%
  summarise(across(starts_with("Heard_"), sum, na.rm = TRUE)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "count") %>%
  mutate(activity = recode(variable, !!!activity_labels),
         type = "Heard About")

```

Warning: There was 1 warning in `summarise()`.

! In argument: `across(starts\_with("Heard\_"), sum, na.rm = TRUE)`.

Caused by warning:

! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.

Supply arguments directly to `.fns` through an anonymous function instead.

Previously

```
across(a:b, mean, na.rm = TRUE)
```

Now

```
across(a:b, \(x) mean(x, na.rm = TRUE))
```

```

part_sums <- df %>%
  summarise(across(starts_with("Part_"), sum, na.rm = TRUE)) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "count") %>%
  mutate(activity = recode(variable, !!!part_labels),
         type = "Participated")

combined <- bind_rows(heard_sums, part_sums) %>%
  mutate(pct = round(count / 182 * 100, 1))

# Grouped bar chart
p3 <- plot_ly(
  combined,
  x      = ~activity,
  y      = ~pct,
  color  = ~type,
  type   = "bar",
  colors = c("Heard About" = "#5B9BD5", "Participated" = "#ED7D31"),

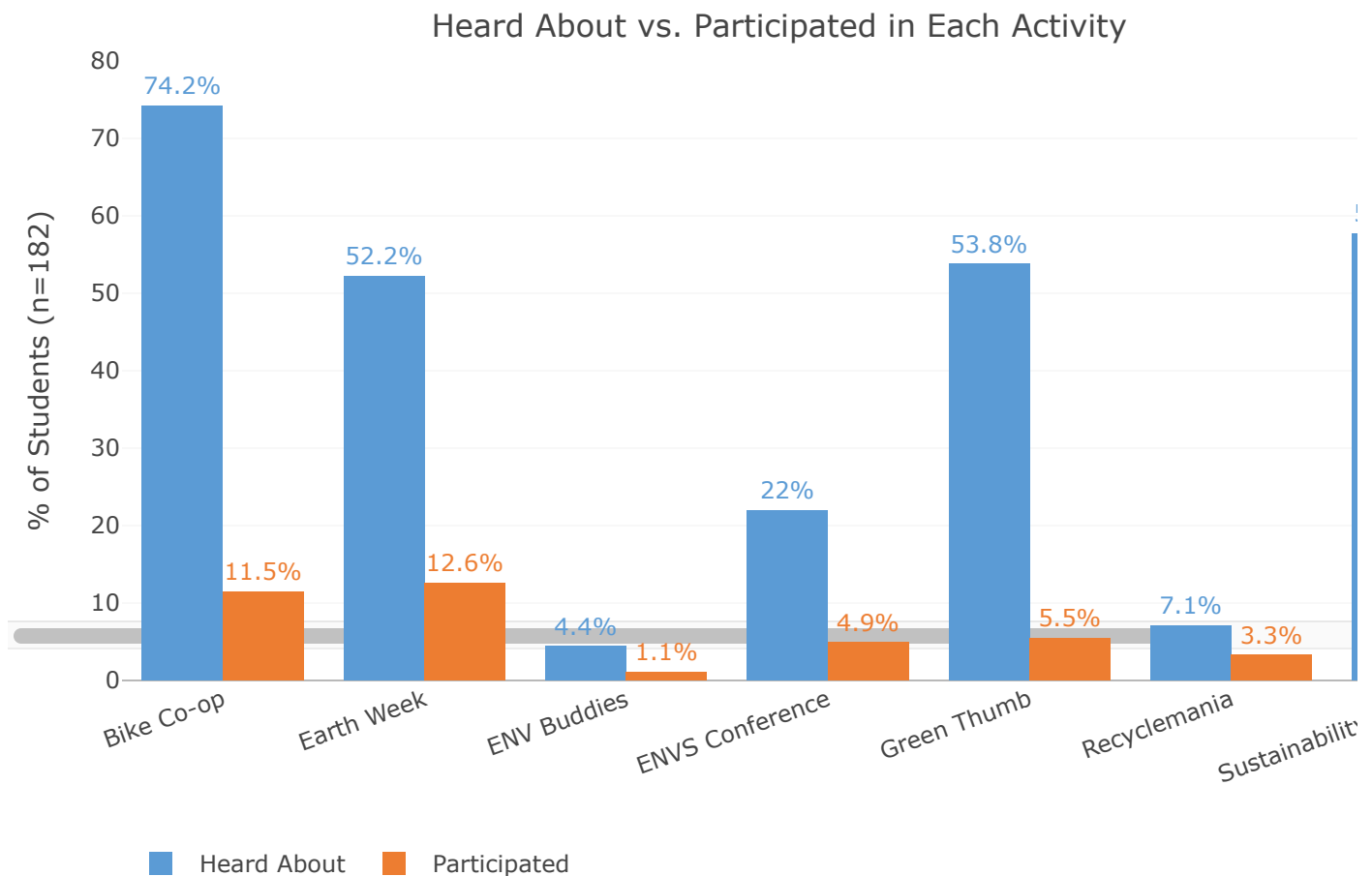
```

```

text      = ~paste0(pct, "%"),
textposition = "outside",
hovertemplate = "<b>{%x}</b><br>{%data.name}: {%y}%<br>({%customdata} students)<extra><
customdata = ~count
) %>%
layout(
  barmode = "group",
  title   = list(text = "Heard About vs. Participated in Each Activity",
                 font = list(size = 16)),
  xaxis   = list(title = "", tickangle = -20),
  yaxis   = list(title = "% of Students (n=182)", range = c(0, 80)),
  legend  = list(orientation = "h", y = -0.25),
  plot_bgcolor = "rgba(0,0,0,0)",
  paper_bgcolor = "rgba(0,0,0,0)"
)

```

p3



Comparison of how many students heard about vs. actually participated in each sustainability activity

Key observation: There is a large gap between *awareness* and *participation* across all activities — students have heard of most activities but very few have actually taken part. This highlights a potential **outreach-to-engagement gap** worth addressing.

3. Confidence Level in Understanding Sustainability

```
# Recode Confidence to ordered factor
conf_levels <- c("Not Confident at All", "Slightly Confident",
                "Neutral", "Confident", "Extremely Confident")

df_conf <- df %>%
  mutate(
    Confidence = factor(Confidence, levels = conf_levels, ordered = TRUE),
    School1 = word(School, 1) # extract first school for breakdown
  )

# --- Plot 4: Overall confidence distribution ---
conf_counts <- df_conf %>%
  count(Confidence) %>%
  mutate(pct = round(n / sum(n) * 100, 1))

conf_colors <- c("#E74C3C", "#E67E22", "#F1C40F", "#2ECC71", "#27AE60")

p4 <- plot_ly(
  conf_counts,
  x = ~Confidence,
  y = ~pct,
  type = "bar",
  marker = list(
    color = conf_colors,
    line = list(color = "white", width = 1.5)
  ),
  text = ~paste0(pct, "%\n(n=", n, ")"),
  textposition = "outside",
  hovertemplate = "<b>{x}</b><br>{y}% of students<br>Count: %{customdata}<extra></extra>"
  customdata = ~n
) %>%
  layout(
    title = list(text = "Overall Confidence in Understanding Sustainability",
                 font = list(size = 16)),
    xaxis = list(title = "Confidence Level", tickangle = -15),
    yaxis = list(title = "% of Students", range = c(0, 45)),
    plot_bgcolor = "rgba(0,0,0,0)",
    paper_bgcolor = "rgba(0,0,0,0)"
  )

# --- Plot 5: Confidence by School (stacked bar) ---
school_labels <- c(
  BUS = "Business", HSE = "Health Sci & Ed",
  IDS = "Undeclared/IDS", SAL = "Arts & Letters",
  SAM = "Science & Math", SCS = "Social & Cultural"
)

conf_school <- df_conf %>%
```

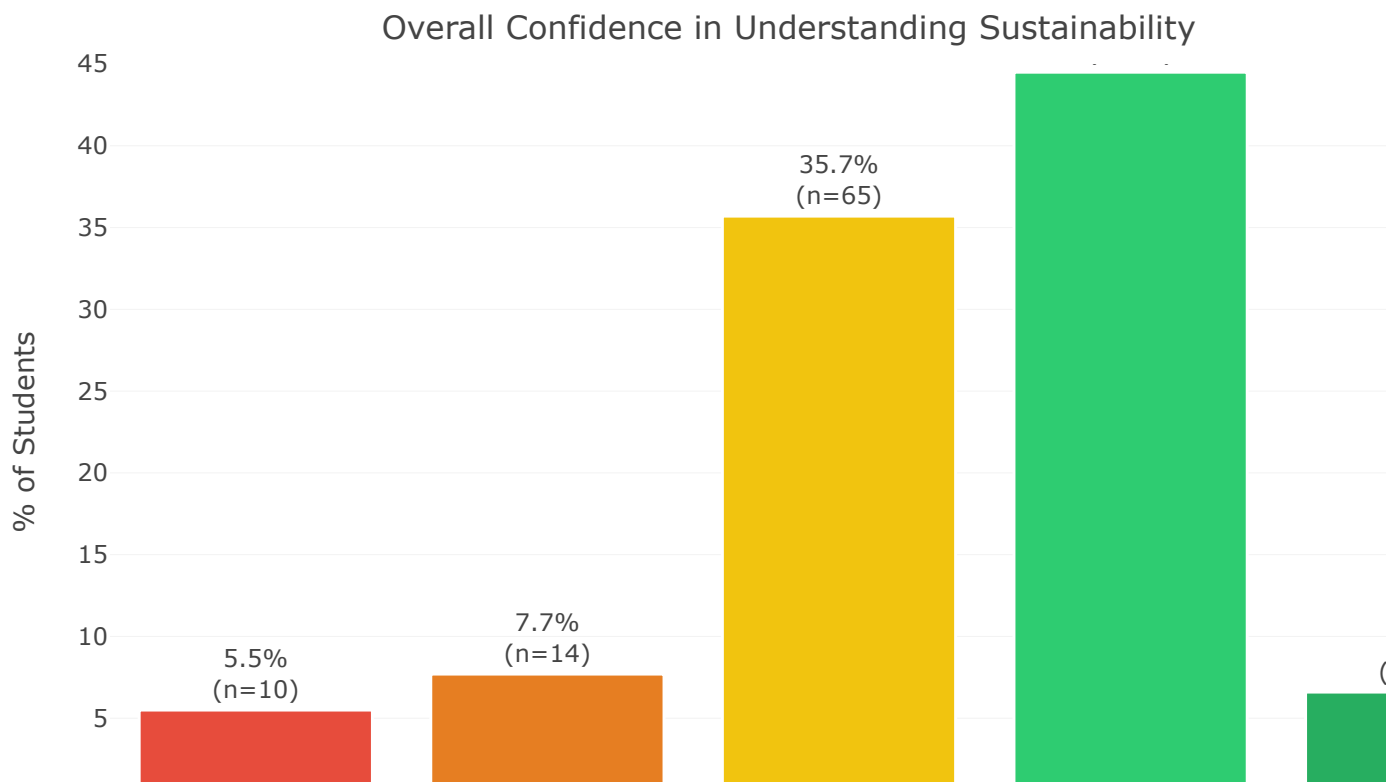
```

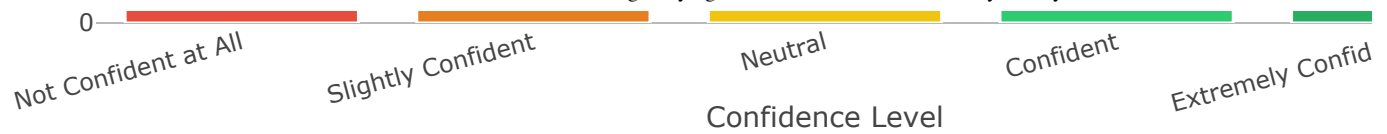
filter(!is.na(School1), !is.na(Confidence)) %>%
mutate(School_Full = recode(School1, !!!school_labels)) %>%
count(School_Full, Confidence) %>%
group_by(School_Full) %>%
mutate(pct = round(n / sum(n) * 100, 1)) %>%
ungroup()

p5 <- plot_ly(
  conf_school,
  x      = ~School_Full,
  y      = ~pct,
  color  = ~Confidence,
  type   = "bar",
  colors = conf_colors,
  hovertemplate = "<b>{%x}</b><br>{%data.name}: {%y}%<extra></extra>"
) %>%
  layout(
    barmode = "stack",
    title   = list(text = "Confidence Level by First-Listed School",
                  font = list(size = 16)),
    xaxis   = list(title = "School", tickangle = -15),
    yaxis   = list(title = "% of Students", ticksuffix = "%"),
    legend  = list(title = list(text = "Confidence Level"),
                  orientation = "h", y = -0.3),
    plot_bgcolor = "rgba(0,0,0,0)",
    paper_bgcolor = "rgba(0,0,0,0)"
  )

p4

```

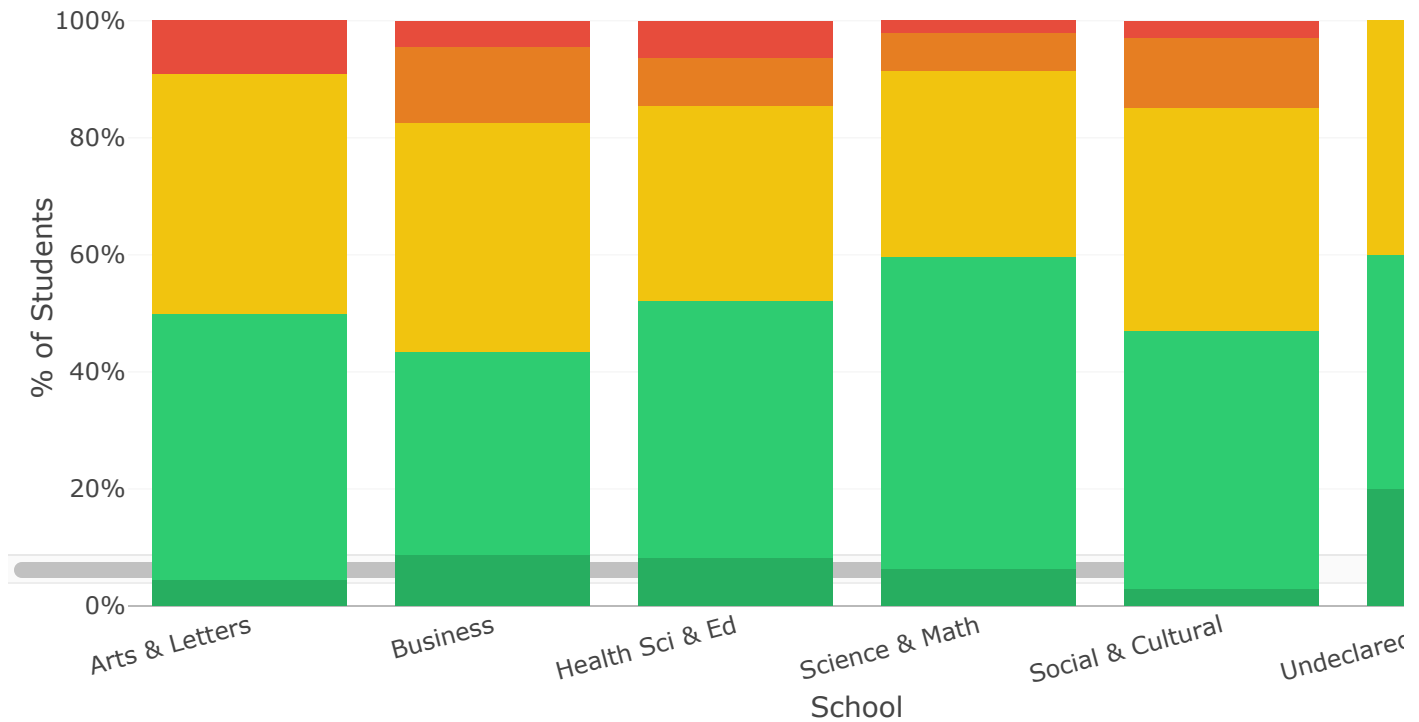




Confidence level distribution overall and broken down by School

p5

Confidence Level by First-Listed School



Confidence Level Not Confident at All Slightly Confident Neutral Confident Extr

Confidence level distribution overall and broken down by School

4. Office Preference: Physical vs. Online

```
# --- Plot 6: Overall office preference ---
office_counts <- df %>%
  filter(!is.na(Office_space)) %>%
  count(Office_space) %>%
  mutate(pct = round(n / sum(n) * 100, 1))

p6 <- plot_ly(
  office_counts,
```

```

labels = ~Office_space,
values = ~n,
type = "pie",
hole = 0.45,
marker = list(
  colors = c("#2E86C1", "#E74C3C", "#95A5A6"),
  line = list(color = "white", width = 2)
),
textinfo = "label+percent",
hovertemplate = "<b>{%label}</b><br>Count: {%value}<br>{%percent}<extra></extra>"
) %>%
layout(
  title = list(text = "Office Preference: Physical vs. Online",
    font = list(size = 16)),
  showlegend = TRUE,
  legend = list(orientation = "h", y = -0.1)
)

# --- Plot 7: Office preference by School ---
office_school <- df %>%
  filter(!is.na(Office_space)) %>%
  separate("School", into = c("School1", "School2", "School3"),
    sep = " ", extra = "drop", fill = "right") %>%
  filter(!is.na(School1)) %>%
  mutate(School_Full = recode(School1, !!!school_labels)) %>%
  count(School_Full, Office_space) %>%
  group_by(School_Full) %>%
  mutate(pct = round(n / sum(n) * 100, 1)) %>%
  ungroup()

office_colors <- c(
  "Physical Office" = "#2E86C1",
  "Online Office" = "#E74C3C",
  "No preference" = "#95A5A6"
)

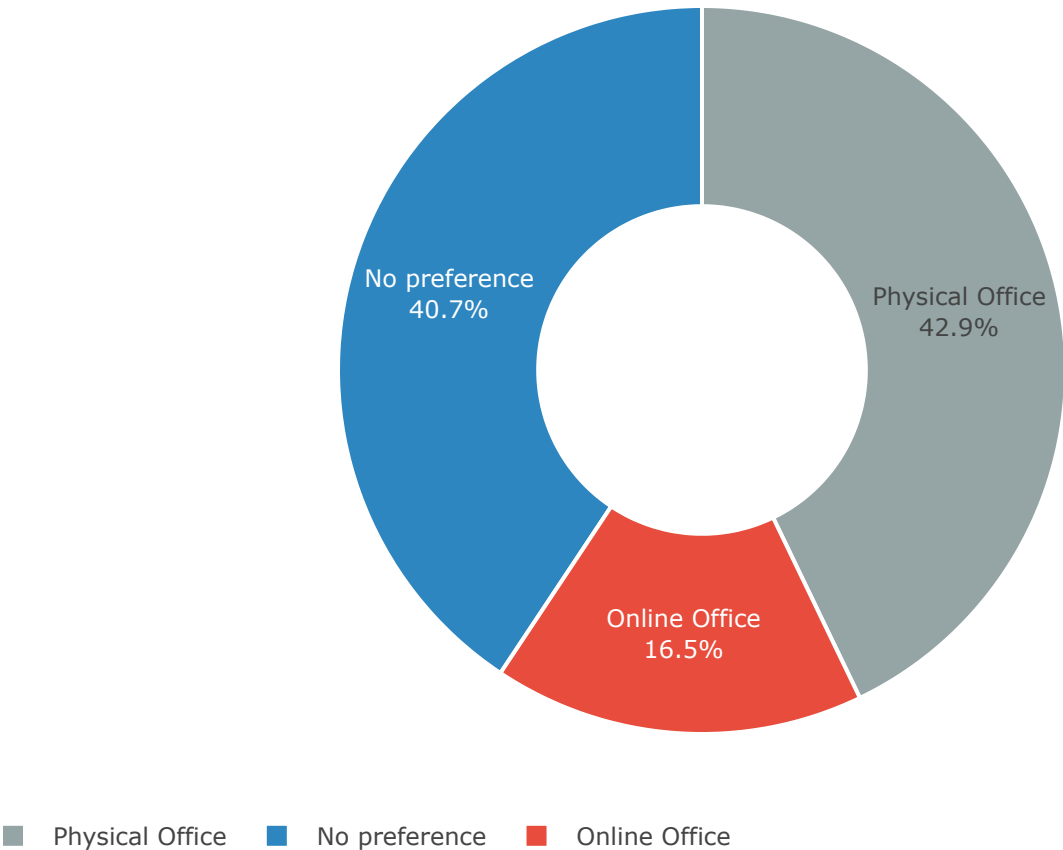
p7 <- plot_ly(
  office_school,
  x = ~School_Full,
  y = ~pct,
  color = ~Office_space,
  type = "bar",
  colors = office_colors,
  hovertemplate = "<b>{%x}</b><br>{%data.name}: {%y}%<extra></extra>"
) %>%
layout(
  barmode = "stack",
  title = list(text = "Office Preference by First-Listed School",
    font = list(size = 16)),
  xaxis = list(title = "School", tickangle = -15),
  yaxis = list(title = "% of Students", ticksuffix = "%"),

```

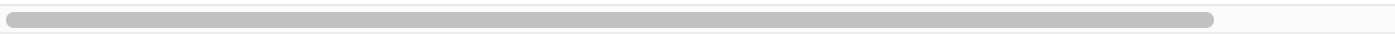
```
legend = list(title = list(text = "Preference"),
              orientation = "h", y = -0.3),
plot_bgcolor = "rgba(0,0,0,0)",
paper_bgcolor = "rgba(0,0,0,0)"
)
```

p6

Office Preference: Physical vs. Online



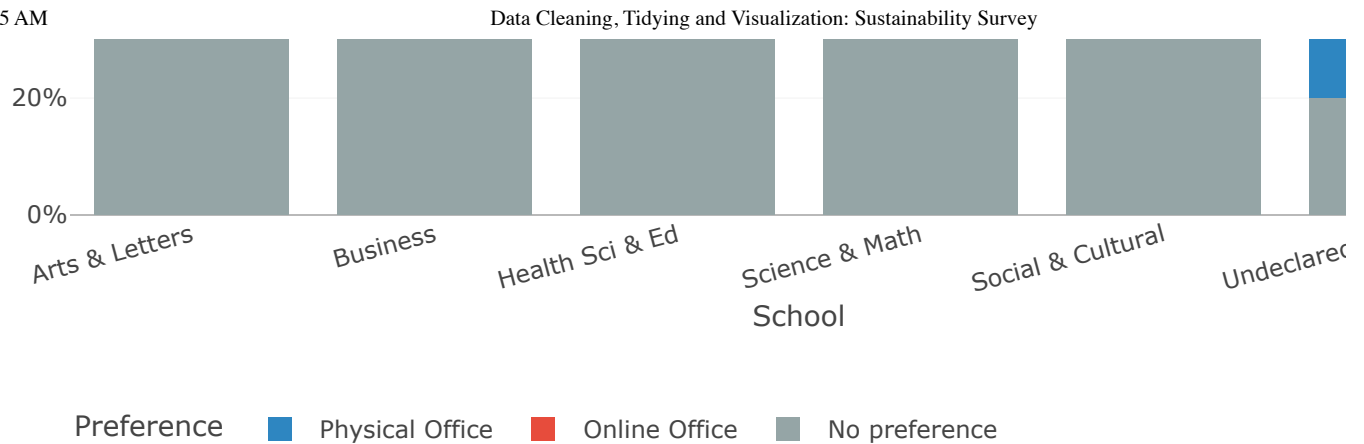
Student preferences for physical vs. online Sustainability Office



p7

Office Preference by First-Listed School





Student preferences for physical vs. online Sustainability Office

5. Statistical Analysis

Chi-square Test: Office Preference vs. School

```
# Does school affiliation influence office preference?
df_chisq <- df %>%
  separate("School", into = c("School1", "School2", "School3"),
           sep = " ", extra = "drop", fill = "right") %>%
  filter(!is.na(School1), !is.na(Office_space))

chisq_result <- chisq.test(table(df_chisq$School1, df_chisq$Office_space))
```

Warning in chisq.test(table(df_chisq\$School1, df_chisq\$Office_space)):
Chi-squared approximation may be incorrect

```
cat("Chi-square Test: Office Preference vs. School\n")
```

Chi-square Test: Office Preference vs. School

```
cat("-----\n")
```

```
cat("Chi-square statistic:", round(chisq_result$statistic, 3), "\n")
```

Chi-square statistic: 7.791

```
cat("Degrees of freedom :", chisq_result$parameter, "\n")
```

Degrees of freedom : 10


```
cat("p-value          :", round(chisq_result$p.value, 4), "\n\n")
```

p-value : 0.6493

```
if (chisq_result$p.value < 0.05) {
  cat("✅ Result: SIGNIFICANT (p < 0.05)\n")
  cat("  School affiliation IS associated with office preference.\n")
} else {
  cat("❌ Result: NOT significant (p >= 0.05)\n")
  cat("  No strong evidence that school affiliation affects office preference.\n")
}
```

❌ Result: NOT significant (p >= 0.05)
No strong evidence that school affiliation affects office preference.

Chi-square Test: Gender vs. Tuition Willingness

```
# Does gender influence willingness to increase tuition?
df_gt <- df %>%
  mutate(
    Gender = case_when(
      Gender == "Female" ~ "Female",
      Gender == "Male" ~ "Male",
      Gender == "Other/ prefer not to say" ~ "Other"
    )
  ) %>%
  filter(!is.na(Gender), !is.na(Increase_rate))

chisq_gt <- chisq.test(table(df_gt$Gender, df_gt$Increase_rate))
```

Warning in chisq.test(table(df_gt\$Gender, df_gt\$Increase_rate)): Chi-squared approximation may be incorrect

```
cat("Chi-square Test: Gender vs. Tuition Willingness\n")
```

Chi-square Test: Gender vs. Tuition Willingness

```
cat("-----\n")
```

```
cat("Chi-square statistic:", round(chisq_gt$statistic, 3), "\n")
```

Chi-square statistic: 14.746

```
cat("Degrees of freedom :", chisq_gt$parameter, "\n")
```

Degrees of freedom : 8

```
cat("p-value          :", round(chisq_gt$p.value, 4), "\n\n")
```

p-value : 0.0643

```
if (chisq_gt$p.value < 0.05) {
  cat("✅ Result: SIGNIFICANT (p < 0.05)\n")
  cat("  Gender IS associated with tuition willingness.\n")
} else {
  cat("❌ Result: NOT significant (p >= 0.05)\n")
  cat("  No strong evidence that gender affects tuition willingness.\n")
}
```

❌ Result: NOT significant (p >= 0.05)
No strong evidence that gender affects tuition willingness.

Confidence Level vs. Tuition Willingness

```
conf_levels <- c("Not Confident at All", "Slightly Confident",
                "Neutral", "Confident", "Extremely Confident")

tuition_levels <- c("Not willing at all", "$1-$10", "$11-$20",
                  "$21-$30", "anything above $30")

# Heatmap: Confidence vs Tuition Willingness
heatmap_df <- df %>%
  filter(!is.na(Confidence), !is.na(Increase_rate)) %>%
  mutate(
    Confidence = factor(Confidence, levels = conf_levels),
    Increase_rate = factor(Increase_rate, levels = tuition_levels)
  ) %>%
  count(Confidence, Increase_rate) %>%
  group_by(Confidence) %>%
  mutate(pct = round(n / sum(n) * 100, 1)) %>%
  ungroup()

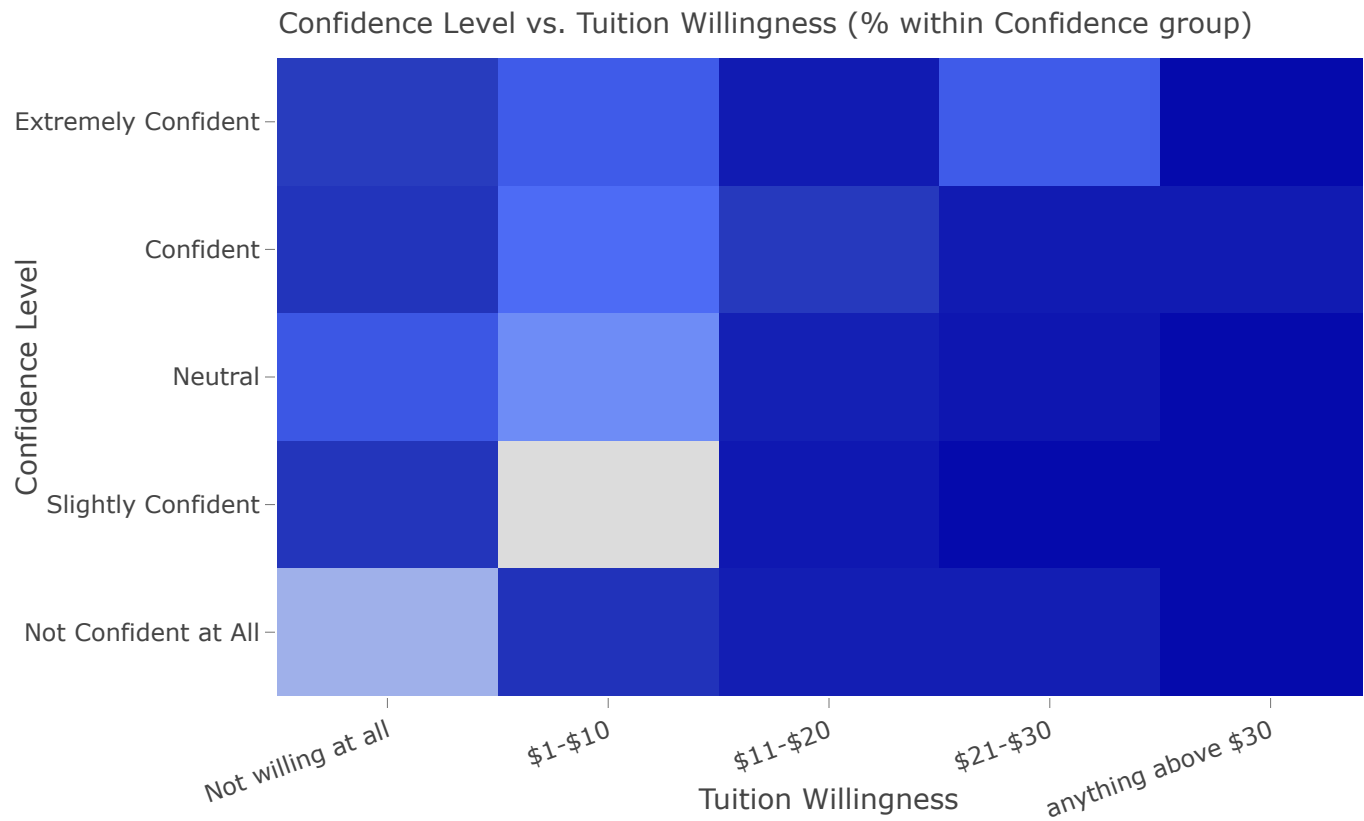
# Pivot to matrix for heatmap
heatmap_matrix <- heatmap_df %>%
  select(Confidence, Increase_rate, pct) %>%
  pivot_wider(names_from = Increase_rate, values_from = pct, values_fill = 0)

mat <- as.matrix(heatmap_matrix[, -1])
rownames(mat) <- heatmap_matrix$Confidence

p8 <- plot_ly(
  x = tuition_levels,
  y = conf_levels,
  z = mat,
```


✔ Significant positive correlation – students who are MORE confident in sustainability tend to be MORE willing to increase tuition.

p8



Relationship between confidence in sustainability understanding and willingness to increase tuition

